

Range

The **Range** (R) is the difference between the largest observation (L) and the smallest observation (S): $R = L - S$.

The **Coefficient of Range** (CR) is a relative measure of dispersion: $CR = \frac{L-S}{L+S}$.

Example:

- Find the range & Coefficient of Range for the following data: 20, 35, 25, 30, 15.

Solution:

The largest value is $L = 35$. The smallest value is $S = 15$.

$$\text{Range}(R) = L - S = 35 - 15 = \mathbf{20}$$

$$\text{Coefficient of Range}(CR) = \frac{L - S}{L + S} = \frac{35 - 15}{35 + 15} = \frac{20}{50} = \mathbf{0.4}$$

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Example:

- Find the range & Coefficient of Range for the following data: 25, 38, 45, 30, 15.

Solution:

The largest value is $L = 45$. The smallest value is $S = 15$.

$$\text{Range}(R) = L - S = 45 - 15 = \mathbf{30}$$

$$\text{Coefficient of Range}(CR) = \frac{L - S}{L + S} = \frac{45 - 15}{45 + 15} = \frac{30}{60} = \mathbf{0.5}$$

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Example:

- Find the range & Coefficient of Range for the following data:

X	1	2	3	4	5	6	7
F	5	9	12	17	14	10	6

Solution:

For a discrete frequency distribution, the range is determined only by the extreme values of X , regardless of their frequencies.

The largest value is $L = 7$. The smallest value is $S = 1$.

$$\text{Range}(R) = L - S = 7 - 1 = 6$$

$$\text{Coefficient of Range}(CR) = \frac{L - S}{L + S} = \frac{7 - 1}{7 + 1} = \frac{6}{8} = 0.75$$

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Example:

- Find the range & Coefficient of Range:

X	0-10	10-20	20-30	30-40	40-50	50-60
F	12	18	27	20	17	6

Solution:

For a grouped frequency distribution, the range is the difference between the upper limit of the last class and the lower limit of the first class.

The largest value is $L = 60$ (Upper limit of 50-60). The smallest value is $S = 0$ (Lower limit of 0-10).

$$\text{Range}(R) = L - S = 60 - 0 = 60$$

$$\text{Coefficient of Range}(CR) = \frac{L - S}{L + S} = \frac{60 - 0}{60 + 0} = \frac{60}{60} = 1.0$$

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Quartile Deviation

The **Interquartile Range** (IQR) is $Q_3 - Q_1$.

The **Quartile Deviation** (QD) is $\frac{Q_3 - Q_1}{2}$.

The **Coefficient of Quartile Deviation** (CQD) is $\frac{Q_3 - Q_1}{Q_3 + Q_1}$.

Example:

- Find interquartile range, quartile deviation and coefficient of quartile deviation: 28, 18, 20, 24, 27, 30, 15.

Solution:

1. **Sort Data:** 15, 18, 20, 24, 27, 28, 30. ($N = 7$)

2. **Find Quartiles:** We use the position formula $\text{Position} = \frac{k(N+1)}{4}$.

- Q_1 position ($\frac{1(7+1)}{4} = 2^{\text{nd}}$ observation) is $Q_1 = 18$.
- Q_3 position ($\frac{3(7+1)}{4} = 6^{\text{th}}$ observation) is $Q_3 = 28$.

3. **Calculate Measures:**

$$\text{Interquartile Range}(IQR) = Q_3 - Q_1 = 28 - 18 = 10$$

$$\text{Quartile Deviation}(QD) = \frac{Q_3 - Q_1}{2} = \frac{10}{2} = 5.0$$

$$\text{Coefficient of Quartile Deviation}(CQD) = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{10}{28 + 18} = \frac{10}{46} \approx 0.2174$$



Example:

- Find interquartile range, quartile deviation and coefficient of quartile deviation:

X	10	20	30	40	50	60
F	2	8	20	35	42	20

Solution:

1. Calculate Cumulative Frequency (*c. f.*):

X	F	c. f.
10	2	2
20	8	10
30	20	30
40	35	65
50	42	107
60	20	127
Total	N = 127	

2. **Find Quartiles:** We use the position formula $\text{Position} = \frac{k(N+1)}{4}$.

- Q_1 position ($\frac{127+1}{4} = 32^{\text{nd}}$). The first *c. f.* ≥ 32 is 65, so $Q_1 = 40$.
- Q_3 position ($3 \times 32 = 96^{\text{th}}$). The first *c. f.* ≥ 96 is 107, so $Q_3 = 50$.

3. Calculate Measures:

$$\text{Interquartile Range}(IQR) = Q_3 - Q_1 = 50 - 40 = 10$$

$$\text{Quartile Deviation}(QD) = \frac{Q_3 - Q_1}{2} = \frac{10}{2} = 5.0$$

$$\text{Coefficient of Quartile Deviation}(CQD) = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{10}{50 + 40} = \frac{10}{90} \approx 0.1111$$



Example:

- Find quartile deviation and coefficient of quartile deviation: 4, 8, 10, 7, 15, 11, 18, 14, 12, 16.

Solution:

1. **Sort Data:** 4, 7, 8, 10, 11, 12, 14, 15, 16, 18. ($N = 10$)

2. **Find Quartiles (Interpolation):**

- Q_1 position ($\frac{1(10+1)}{4} = 2.75^{\text{th}}$).

$$Q_1 = x_2 + 0.75(x_3 - x_2) = 7 + 0.75(8 - 7) = \mathbf{7.75}$$

- Q_3 position ($\frac{3(10+1)}{4} = 8.25^{\text{th}}$).

$$Q_3 = x_8 + 0.25(x_9 - x_8) = 15 + 0.25(16 - 15) = \mathbf{15.25}$$

3. Calculate Measures:

$$\text{Quartile Deviation}(QD) = \frac{Q_3 - Q_1}{2} = \frac{15.25 - 7.75}{2} = \frac{7.5}{2} = \mathbf{3.75}$$

$$\text{Coefficient of Quartile Deviation}(CQD) = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{7.5}{15.25 + 7.75} = \frac{7.5}{23} \approx \mathbf{0.3261}$$



Example:

- Find quartile deviation and coefficient of quartile deviation:

X	20	40	60	80	100
f	4	10	15	20	11

Solution:

1. Calculate Cumulative Frequency (*c. f.*):

X	f	c. f.
20	4	4
40	10	14
60	15	29
80	20	49
100	11	60
Total	N = 60	

2. Find Quartiles:

- Q_1 position ($\frac{60+1}{4} = 15.25^{\text{th}}$). The first $c. f. \geq 15.25$ is 29, so $Q_1 = 60$.
- Q_3 position ($3 \times 15.25 = 45.75^{\text{th}}$). The first $c. f. \geq 45.75$ is 49, so $Q_3 = 80$.

3. Calculate Measures:

$$\text{Quartile Deviation}(QD) = \frac{Q_3 - Q_1}{2} = \frac{80 - 60}{2} = \frac{20}{2} = 10.0$$

$$\text{Coefficient of Quartile Deviation}(CQD) = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{20}{80 + 60} = \frac{20}{140} = \frac{1}{7} \approx 0.1429$$

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Mean Deviation

The **Mean Deviation (M.D.)** is the arithmetic mean of the absolute deviations of the observations from an average (Mean, Median, or Mode).

M.D. Formulae:

$$\text{M.D. from } A = \frac{\sum |X - A|}{N} \text{ (Ungrouped)}$$

$$\text{M.D. from } A = \frac{\sum f|X - A|}{N} \text{ (Frequency Distribution)}$$

$$\text{Coefficient of M.D.} = \frac{\text{M.D.}}{\text{Average } (A)}$$

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Mean Deviation

The **Mean Deviation (M.D.)** is the arithmetic mean of the absolute deviations of the observations from an average (Mean or Median).

M.D. Formulae:

$$\text{M.D. from } A = \frac{\sum |X - A|}{N}$$

$$\text{Coefficient of M.D.} = \frac{\text{M.D.}}{\text{Average } (A)}$$



Example:

- Calculate M.D. from Mean & Median & coefficient of Mean Deviation from the following data: 20, 22, 25, 38, 40, 50, 65, 70, 75.

Solution:

1. Calculate Averages:

Data ($N = 9$): 20, 22, 25, 38, **40**, 50, 65, 70, 75.

$$\text{Mean } (\bar{X}) = \frac{\sum X}{N} = \frac{405}{9} = \mathbf{45.0}$$

$$\text{Median } (M) = \left(\frac{9 + 1}{2} \right)^{\text{th}} \text{ item} = 5^{\text{th}} \text{ item} = \mathbf{40.0}$$

2. Calculate Deviations and Sums:

Data Value (X)	Absolute Deviation from Mean ($X - 45$)	Absolute Deviation from Median ($X - 40$)
20	25	20
22	23	18
25	20	15
38	7	2
40	5	0
50	5	10
65	20	25
70	25	30
75	30	35
Total Sum		

3. Calculate Mean Deviation & Coefficient (from Mean):

- $$\text{M.D. from Mean} = \frac{\sum |X - \bar{X}|}{N} = \frac{160}{9} \approx \mathbf{17.78}$$
- $$\text{Coeff. M.D. (Mean)} = \frac{\text{M.D. from Mean}}{\bar{X}} = \frac{17.78}{45.0} \approx \mathbf{0.395}$$

4. Calculate Mean Deviation & Coefficient (from Median):

- $$\text{M.D. from Median} = \frac{\sum |X - M|}{N} = \frac{155}{9} \approx \mathbf{17.22}$$
- $$\text{Coeff. M.D. (Median)} = \frac{\text{M.D. from Median}}{M} = \frac{17.22}{40.0} \approx \mathbf{0.431}$$

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Example:

- Calculate M.D. from Mean & Median & coefficient of Mean Deviation from the following data:

X	20	30	40	50	60	70
F	8	12	20	10	6	4

Solution:

1. Averages:

X	F	FX	c. f.
20	8	160	8
30	12	360	20
40	20	800	40
50	10	500	50

X	F	FX	$c. f.$
60	6	360	56
70	4	280	60
Total	N = 60	$\sum FX = 2460$	

$$\text{Mean } (\bar{X}) = \frac{2460}{60} = \mathbf{41.0}$$

$$\text{Median } (M) : \text{Pos.} = \frac{60 + 1}{2} = 30.5^{\text{th}}. \text{ c. f. } \geq 30.5 \text{ is } 40, \text{ so } M = \mathbf{40.0}$$

2. Deviations:

X	F	$X - \bar{X}$	$F(X - \bar{X})$	$X - M$	$F(X - M)$
20	8	21	168	20	160
30	12	11	132	10	120
40	20	1	20	0	0
50	10	9	90	10	100
60	6	19	114	20	120
70	4	29	116	30	120
Total	60		640		620

3. Mean Deviation & Coefficient:

$$\text{M.D. from Mean} = \frac{640}{60} \approx \mathbf{10.67}$$

$$\text{Coeff. M.D. (Mean)} = \frac{10.67}{41.0} \approx \mathbf{0.260}$$

$$\text{M.D. from Median} = \frac{620}{60} \approx \mathbf{10.33}$$

$$\text{Coeff. M.D. (Median)} = \frac{10.33}{40.0} \approx \mathbf{0.258}$$

Example:

- Calculate the Mean Deviation & coefficient of Mean Deviation from the following data:

Marks	5	15	25	35	45	55
No. of students (F)	10	20	30	40	50	30

Solution:**1. Averages:**

X	F	FX	$c. f.$
5	10	50	10
15	20	300	30
25	30	750	60
35	40	1400	100
45	50	2250	150
55	30	1650	180
Total	$N = 180$	$\sum FX = 6400$	

$$\text{Mean } (\bar{X}) = \frac{6400}{180} \approx \mathbf{35.56}$$

$$\text{Median } (M) : \text{Pos.} = \frac{180 + 1}{2} = 90.5^{\text{th}}. \text{ } c. f. \geq 90.5 \text{ is } 100, \text{ so } M = \mathbf{35.0}$$

2. Deviations from Mean ($\bar{X} = 35.56$):

X	F	$X - 35.56$	$F(X - 35.56)$
5	10	30.56	305.6
15	20	20.56	411.2
25	30	10.56	316.8
35	40	0.56	22.4
45	50	9.44	472.0
55	30	19.44	583.2
Total	180		2111.2

3. Mean Deviation & Coefficient (from Mean):

$$\text{M.D. from Mean} = \frac{2111.2}{180} \approx \mathbf{11.73}$$

$$\text{Coeff. M.D. (Mean)} = \frac{11.73}{35.56} \approx \mathbf{0.330}$$

(The calculation for M.D. from Median is also provided in the tool output: M.D. from Median ≈ 11.67 , Coeff. ≈ 0.333)



Example:

- Calculate the Mean Deviation & coefficient of Mean Deviation from the following data:

Marks	0-10	10-20	20-30	30-40	40-50	50-60
No. of students (F)	10	20	30	50	40	30

Solution:

1. Averages:

Class	F	Midpoint (m)	Fm	$c. f.$
0-10	10	5	50	10
10-20	20	15	300	30
20-30	30	25	750	60
30-40	50	35	1750	110
40-50	40	45	1800	150
50-60	30	55	1650	180
Total	N = 180		$\sum Fm = 6300$	

$$\text{Mean } (\bar{X}) = \frac{6300}{180} = \mathbf{35.0}$$

2. Deviations from Mean ($\bar{X} = 35.0$):

m	F	$m - 35$	$F(m - 35)$
5	10	30	300
15	20	20	400
25	30	10	300
35	50	0	0
45	40	10	400
55	30	20	600
Total	180		2000

3. Mean Deviation & Coefficient (from Mean):

$$\text{M.D. from Mean} = \frac{2000}{180} \approx \mathbf{11.11}$$

$$\text{Coeff. M.D. (Mean)} = \frac{11.11}{35.0} \approx \mathbf{0.317}$$

(Note: If calculated from the Median $M = 37.5$ (from tool output), $M.D. \approx 11.94$, $\text{Coeff.} \approx 0.319$)

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Example:

- Write short notes on:
 - * Range
 - * Interquartile range
 - * Mean deviation
 - * Standard deviation
 - * Variance

Solution:

Short Notes on Measures of Dispersion

- **Range:** The simplest measure of dispersion, defined as the difference between the **largest observation** (L) and the **smallest observation** (S) in a dataset: $R = L - S$. It is easy to calculate but relies solely on the two extreme values, making it highly susceptible to outliers.
- **Interquartile Range (IQR):** A measure of statistical dispersion equal to the difference between the third quartile (Q_3) and the first quartile (Q_1): $IQR = Q_3 - Q_1$. It measures the spread of the middle 50% of the data, making it a robust measure that is unaffected by extreme outliers.
- **Mean Deviation (M.D.):** The arithmetic mean of the **absolute deviations** of the observations from a central value (Mean, Median, or Mode): $M.D. = \frac{\sum |X - A|}{N}$. Since it uses absolute values, it mathematically lacks certain desirable properties compared to the standard deviation, but it is conceptually easier to understand.
- **Standard Deviation (S.D.):** The most widely used measure of dispersion, defined as the **square root of the variance**. It measures the typical amount of variation or dispersion of a set of values from the mean, and is expressed in the original units of measurement. The formula is $\sigma = \sqrt{\frac{\sum (X - \bar{X})^2}{N}}$ (for population).
- **Variance:** The mean of the **squared deviations** from the arithmetic mean. It is the square of the standard deviation (σ^2). It is mathematically preferred because squaring deviations removes the negative signs and weighs larger deviations more heavily, but its unit is the square of the original data unit.



Example:

- Explain the measures of dispersion and also find the range & Coefficient of Range for the following data: 20, 35, 25, 30, 15.

Solution:

Explanation of Measures of Dispersion

Measures of Dispersion (or Measures of Variation) quantify the extent to which the values in a dataset are spread out or scattered around a central value. They indicate how representative the central tendency (like the mean or median) is of the data. A low measure of dispersion indicates that the data points tend to be close to the average, while a high measure indicates the data points are spread farther apart. Common measures include Range, Quartile Deviation, Mean Deviation, and Standard Deviation.

Range and Coefficient of Range Calculation

Data: 20, 35, 25, 30, 15.

The largest value is $L = 35$. The smallest value is $S = 15$.

$$\text{Range}(R) = L - S = 35 - 15 = \mathbf{20}$$

$$\text{Coefficient of Range}(CR) = \frac{L - S}{L + S} = \frac{35 - 15}{35 + 15} = \frac{20}{50} = \mathbf{0.4}$$

Variance and Standard Deviation

The **Variance** (σ^2) measures the average squared deviation from the mean, and the **Standard Deviation** (σ) is the square root of the variance, returning the dispersion measurement to the original units.

Key Formulae:

$$\text{Variance } (\sigma^2) = \frac{\sum(X - \bar{X})^2}{N} \quad \text{or} \quad \frac{\sum f(X - \bar{X})^2}{N}$$

$$\text{Standard Deviation } (\sigma) = \sqrt{\text{Variance}}$$

σ

Example:

- Find the Variance and standard deviation for the following individual series: 3, 6, 8, 10, 18.

Solution:

1. Calculate Mean ($\bar{X}$):

$$N = 5$$

$$\bar{X} = \frac{3 + 6 + 8 + 10 + 18}{5} = \frac{45}{5} = \mathbf{9.0}$$

2. Calculate Deviations and Sum of Squares:

X	$X - \bar{X} = X - 9$	$(X - \bar{X})^2$
3	-6	36
6	-3	9
8	-1	1
10	1	1
18	9	81
Sum	0	$\sum(X - \bar{X})^2 = 128$

3. Calculate Variance and Standard Deviation:

$$\text{Variance } (\sigma^2) = \frac{128}{5} = \mathbf{25.6}$$

$$\text{Standard Deviation } (\sigma) = \sqrt{25.6} \approx \mathbf{5.06}$$

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Example:

- Find the variance and standard deviation for the following frequency distribution:

Marks	5-15	15-25	25-35	35-45	45-55	55-65
No. of students (F)	10	20	25	20	15	10

Solution:

1. Calculate Mean ($\bar{X}$):

$$N = \sum F = 100.$$

Class	F	Midpoint (m)	Fm
5-15	10	10	100
15-25	20	20	400
25-35	25	30	750
35-45	20	40	800
45-55	15	50	750
55-65	10	60	600
Total	100		$\sum Fm = 3400$

$$\bar{X} = \frac{3400}{100} = 34.0$$

2. Calculate Weighted Sum of Squared Deviations:

m	F	$m - \bar{X} = m - 34$	$(m - \bar{X})^2$	$F(m - \bar{X})^2$
10	10	-24	576	5760
20	20	-14	196	3920
30	25	-4	16	400
40	20	6	36	720
50	15	16	256	3840
60	10	26	676	6760
Total	100			21400

3. Calculate Variance and Standard Deviation:

$$\text{Variance } (\sigma^2) = \frac{21400}{100} = 214.0$$

$$\text{Standard Deviation } (\sigma) = \sqrt{214} \approx 14.63$$

σ

Example:

- Calculate standard deviation and variance of the given data:

Size (X)	6	7	8	9	10	11	12
Frequency (F)	3	6	9	13	8	5	4

Solution:

1. Calculate Mean ($\bar{X}$):

$$N = \sum F = 48.$$

X	F	FX
6	3	18
7	6	42
8	9	72
9	13	117
10	8	80
11	5	55
12	4	48
Total	48	$\sum FX = 432$

$$\bar{X} = \frac{432}{48} = 9.0$$

2. Calculate Weighted Sum of Squared Deviations:

X	F	$X - \bar{X} = X - 9$	$(X - \bar{X})^2$	$F(X - \bar{X})^2$
6	3	-3	9	27
7	6	-2	4	24
8	9	-1	1	9
9	13	0	0	0
10	8	1	1	8
11	5	2	4	20
12	4	3	9	36
Total	48			124

3. Calculate Variance and Standard Deviation:

$$\text{Variance } (\sigma^2) = \frac{124}{48} \approx \mathbf{2.58}$$

$$\text{Standard Deviation } (\sigma) = \sqrt{2.5833} \approx \mathbf{1.61}$$

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Example:

- Calculate the standard deviation of 100 students:

Mass (kg)	60-62	63-65	66-68	69-71	72-74
No. of students (F)	10	30	20	50	40

Solution:

The classes are **discontinuous** (e.g., 60-62, 63-65). We first convert them to continuous classes (e.g., 59.5-62.5) to find accurate midpoints and apply the formula.

$N = \sum F = \mathbf{150}$. (Note: We use $N = 150$ for calculation, as per the table, rather than the stated $N = 100$.)

1. Calculate Mean ($\bar{X}$):

Class (Cont.)	F	Midpoint (m)	Fm
59.5-62.5	10	61	610
62.5-65.5	30	64	1920
65.5-68.5	20	67	1340
68.5-71.5	50	70	3500
71.5-74.5	40	73	2920
Total	150		$\sum Fm = \mathbf{10290}$

$$\bar{X} = \frac{10290}{150} = \mathbf{68.6}$$

2. Calculate Weighted Sum of Squared Deviations:

m	F	$m - \bar{X} = m - 68.6$	$(m - \bar{X})^2$	$F(m - \bar{X})^2$
61	10	-7.6	57.76	577.6
64	30	-4.6	21.16	634.8
67	20	-1.6	2.56	51.2
70	50	1.4	1.96	98.0
73	40	4.4	19.36	774.4
Total	150			2136.0

3. Calculate Variance and Standard Deviation:

$$\text{Variance } (\sigma^2) = \frac{2136.0}{150} \approx \mathbf{14.24}$$

$$\text{Standard Deviation } (\sigma) = \sqrt{14.24} \approx \mathbf{3.77}$$

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Example:

- Sum of squares of items 2430, mean is 7, N=12, find the variance.
 - * 176.5
 - * 12.38
 - * 153.26
 - * 14

Solution:

The variance is calculated using the raw moments formula:

$$\text{Variance } (\sigma^2) = \frac{\sum X^2}{N} - (\bar{X})^2$$

Given:

- $\sum X^2 = 2430$
- $\bar{X} = 7$
- $N = 12$

$$\sigma^2 = \frac{2430}{12} - (7)^2$$

$$\sigma^2 = 202.5 - 49$$

$$\sigma^2 = \mathbf{153.5}$$

The calculated variance is 153.5. Among the given options, 153.26 is the closest, likely representing a slight rounding error in the option list.